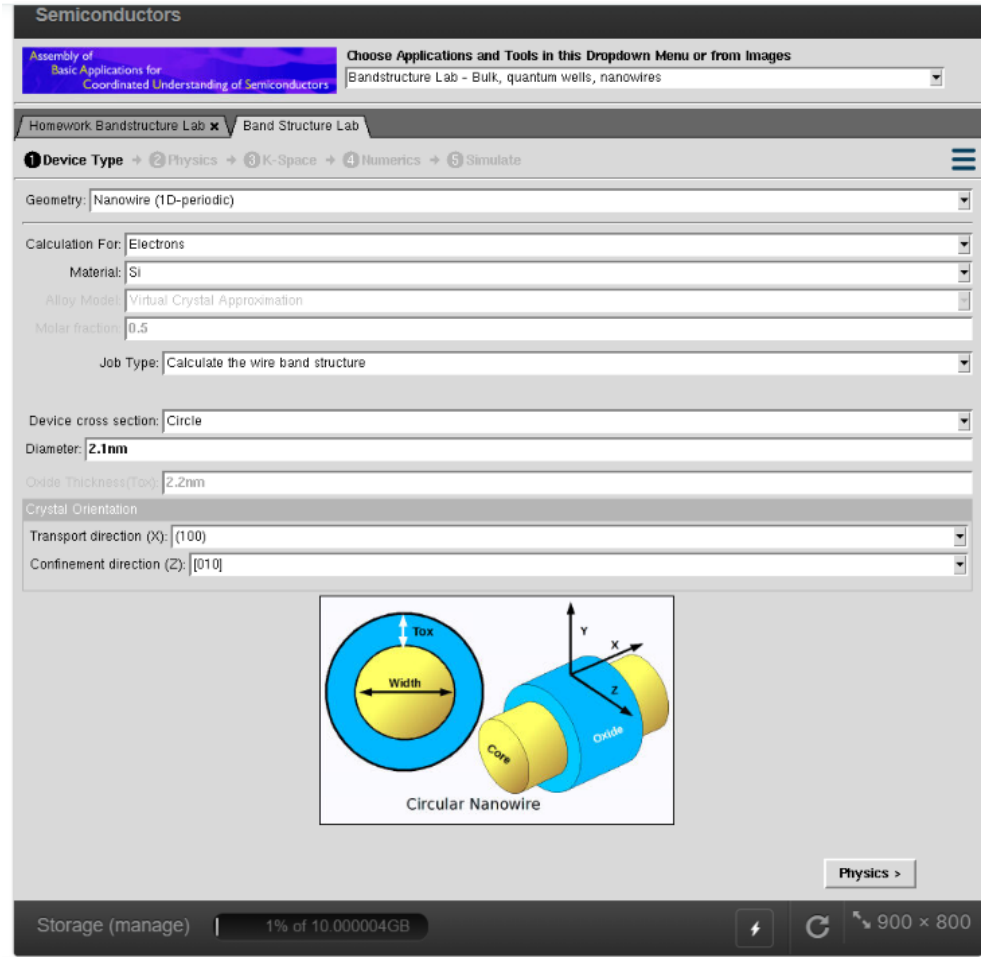
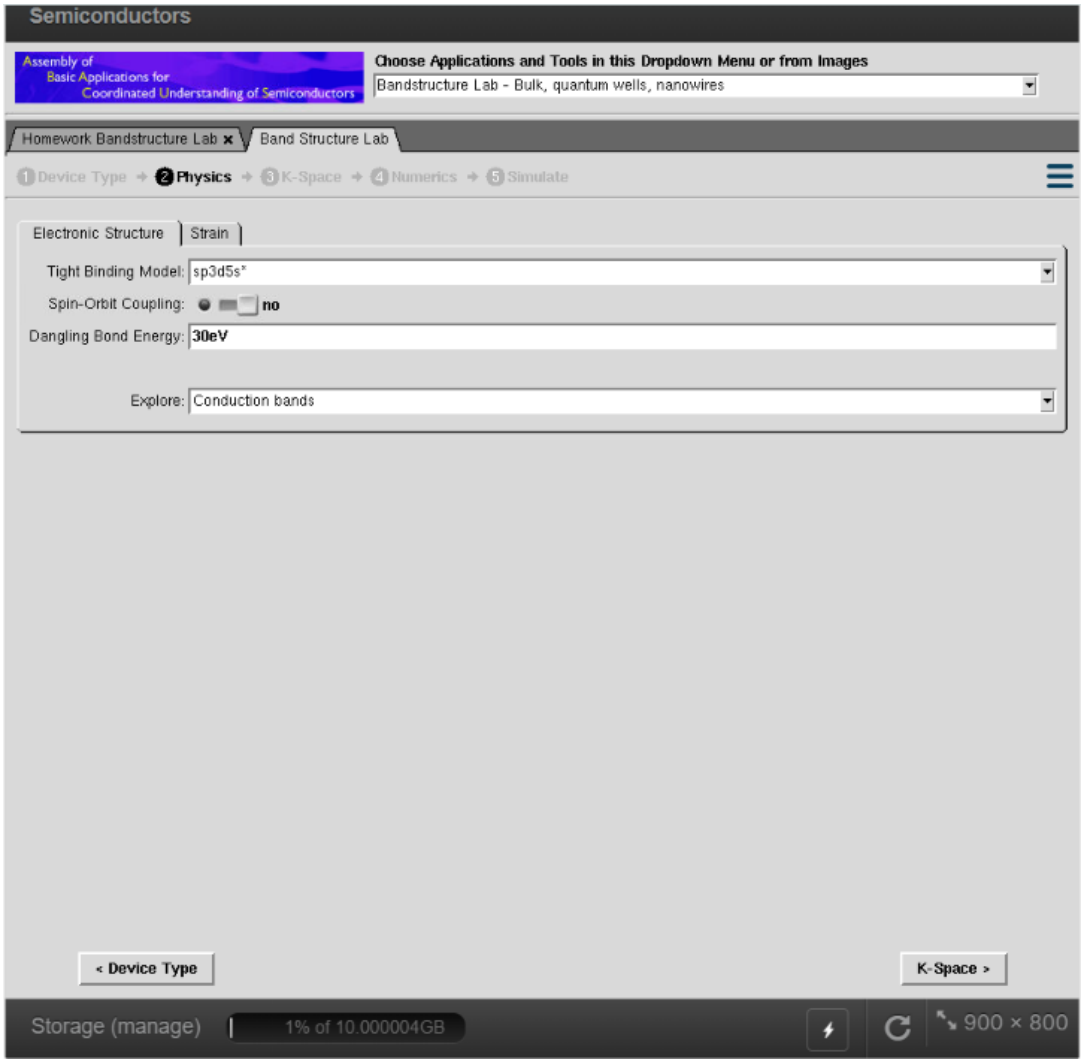
Exp. 5. Band Structure Simulation for Graphene and CNTs using NanoHub

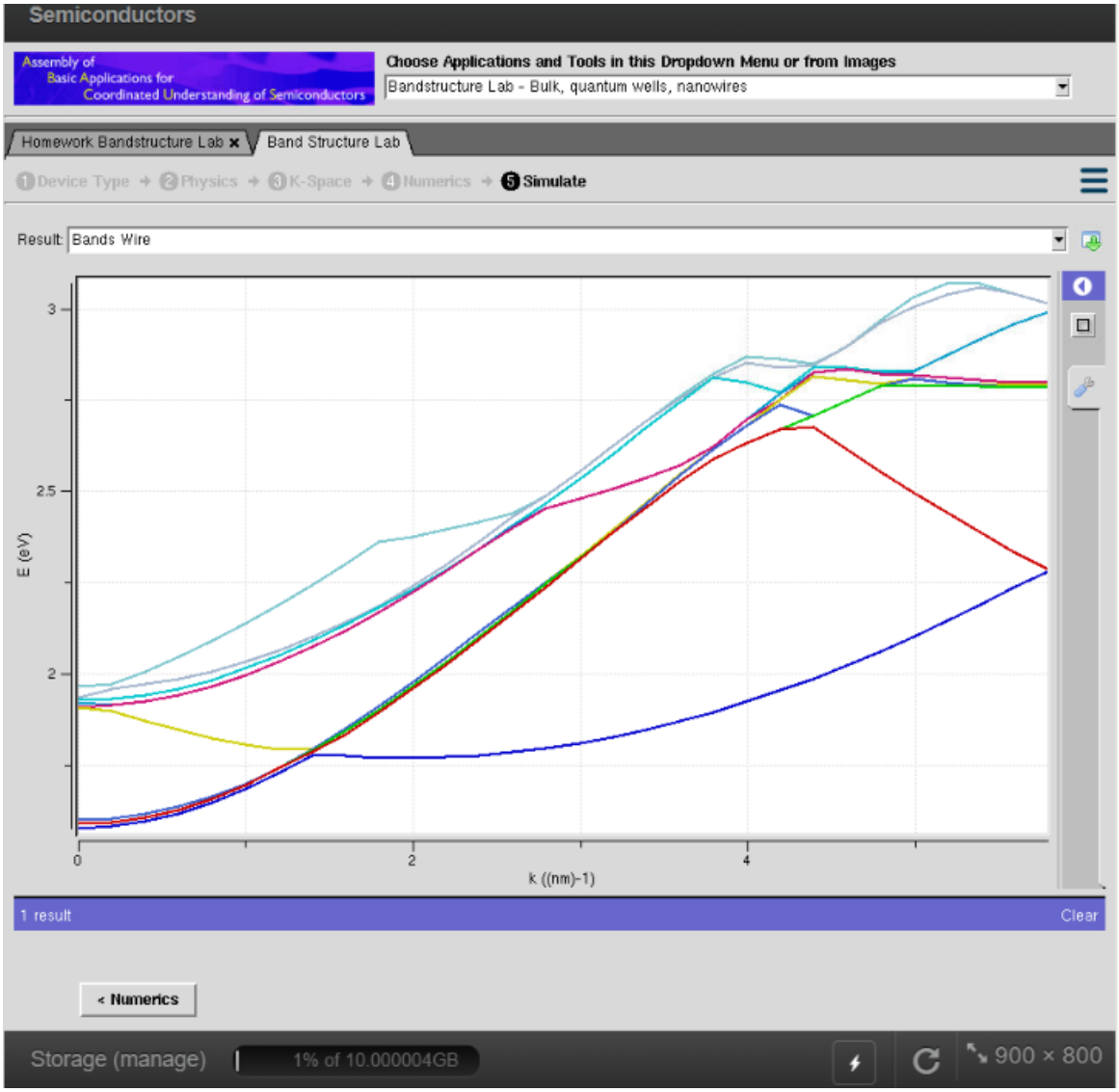
objective:

To simulate the band structure of nanowire for various materials.

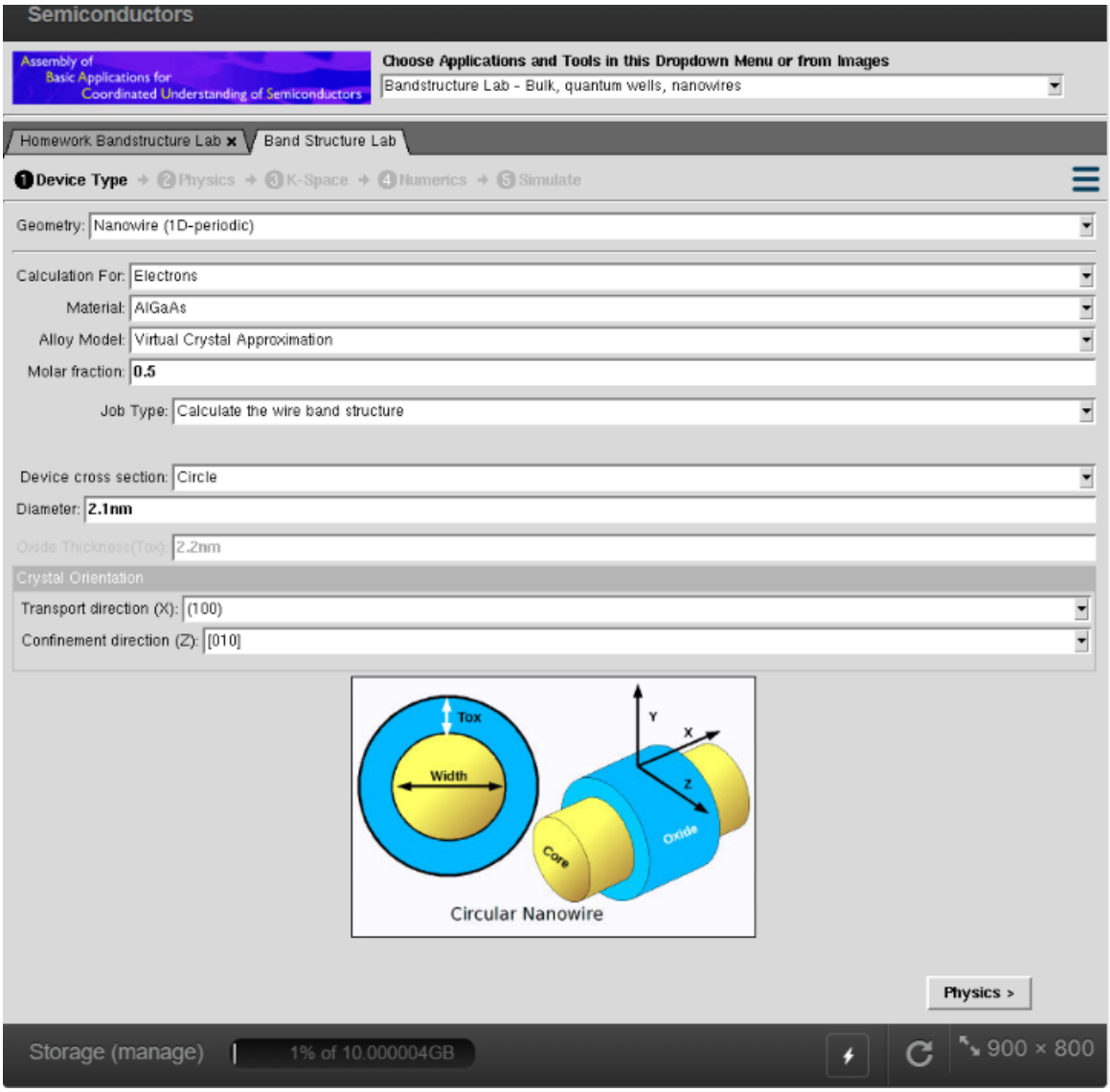
Case Study 1: Simulate the band-structure of InAs circular nanowire

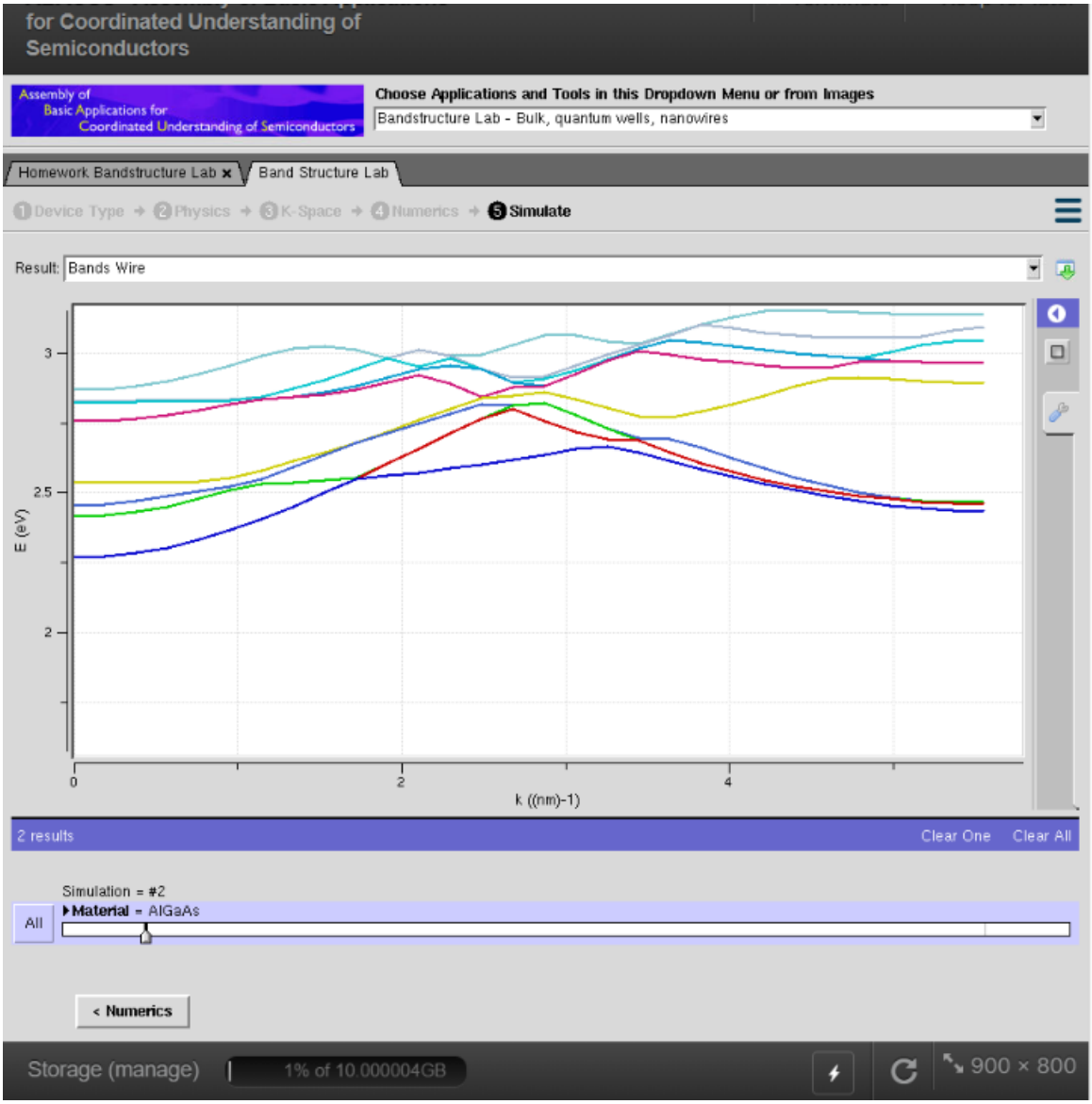



Semiconductors
Assembly of Basic Applications for Coordinated Understanding of Semiconductors
Choose Applications and Tools in this Dropdown Menu or from Images
Bandstructure Lab - Bulk, quantum wells, nanowires
Homework Bandstructure Lab
Band Structure Lab
1 Device Type → 2 Physics → 3 K-Space → 4 Numerics → 5 Simulate
Electronic Structure
Strain
Tight Binding Model: sp3d5s*
Spin-Orbit Coupling: no
Dangling Bond Energy: 30eV
Explore: Conduction bands
« Device Type
K-Space »
Storage (manage)
1% of 10.000004GB
900 × 800


Semiconductors
Assembly of
Basic Applications for
Coordinated Understanding of Semiconductors
Choose Applications and Tools in this Dropdown Menu or from Images
Bandstructure Lab - Bulk, quantum wells, nanowires
Homework Bandstructure Lab
Band Structure Lab
1 Device Type
2 Physics
3 K-Space
4 Numerics
5 Simulate
Result: Bands Wire
E (eV)
3
2.5
2
0
2
4
k ((nm)-1)
1 result
Clear
< Numerics
Storage (manage)
1% of 10.000004GB
900 × 800

Case Study 2: Simulate the band-structure of AlGaAs circular nanowire





Reflections:

- **simulated** periodic 1D nanowire band structures across different material models using the ABACUS tool on nanoHUB.
- **studied** the emergence of transverse energy quantization that occurs when material dimensions scale down to nanometer limits.
- **compared** how material adjustments and model selections alter conduction paths, helping optimize wires for nanoelectronic transport.